

RAFFLES INSTITUTION

2014 Year 6 Preliminary Examination
Higher 2

BIOLOGY

9648/01

Paper 1 Multiple Choice

29th September 2014

1 hour 15 min

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write your name and shade your Index Number on the Answer Sheet in the spaces provided unless this has been done for you.

There are **forty** questions in this paper. Answer all questions. For each question there are four possible answers **A, B, C, and D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

Calculators may be used.

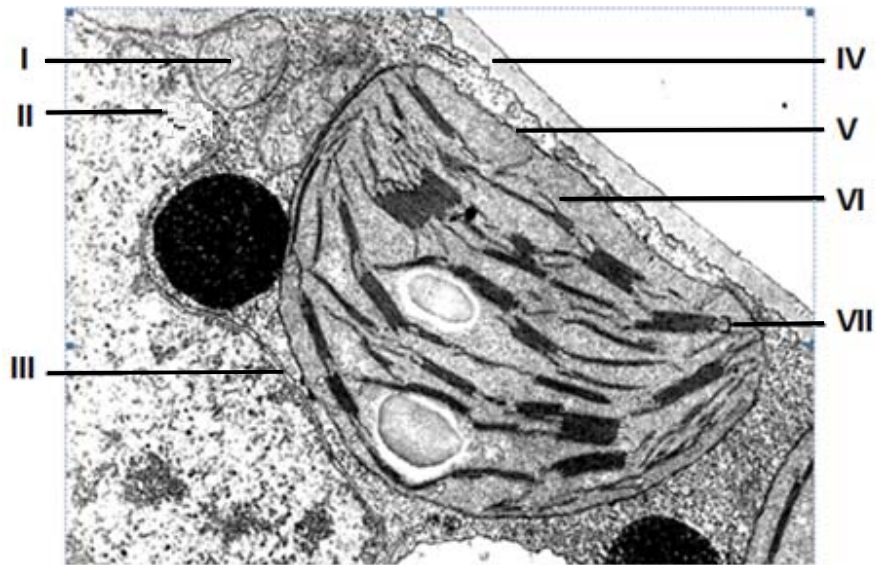
(Erase all mistakes completely. Do not bend or fold the OMR Answer Sheet).

This document consists of **26** printed pages.



Raffles Institution
Internal Examination

1. The diagram below shows part of a eukaryotic cell.



Which of the structures labelled I to VII are correctly matched to the structures/processes in the table below?

	composed of two membranes	site of Calvin cycle	composed of cellulose	stack of membrane bound structures	site of ATP synthesis
A	III	VI	IV	VII	I
B	III	IV	VI	V	I
C	V	VI	IV	VII	II
D	V	VI	III	I	II

2. A student investigating the relative carbohydrate concentrations of different brands of strawberry jam obtained the following results.

Which brand of jam contained the lowest concentration of sucrose?

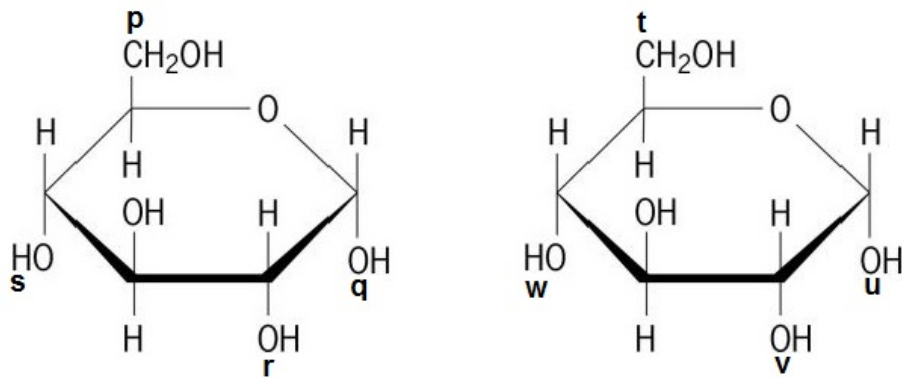
brand of jam	total reducing sugar content per 100g of jam (arbitrary units)	
	untreated jam	jam after hydrolysis with invertase
A	54	76
B	60	84
C	73	93
D	43	65

3. The table describes some molecules found in a cell surface membrane.

Which of the following is correct?

	cholesterol	glycolipid	glycoprotein	phospholipid	protein
A	is a precursor of certain hormones	involved in active transport	transports substances across the membrane	is amphipathic in nature	involved in DNA replication
B	regulates membrane fluidity at different temperatures	involved in cell-cell recognition	increases membrane fluidity at high temperature	involved in DNA replication	limits the permeability of the membrane
C	decreases membrane fluidity at high temperature	involved in cell-cell recognition	acts as a signal binding site during cell signaling	limits the permeability of the membrane	can change in conformation when phosphorylated by ATP
D	has a four ring structure	involved in active transport	acts as a ligand binding site	is amphoteric in nature	limits the permeability of the membrane

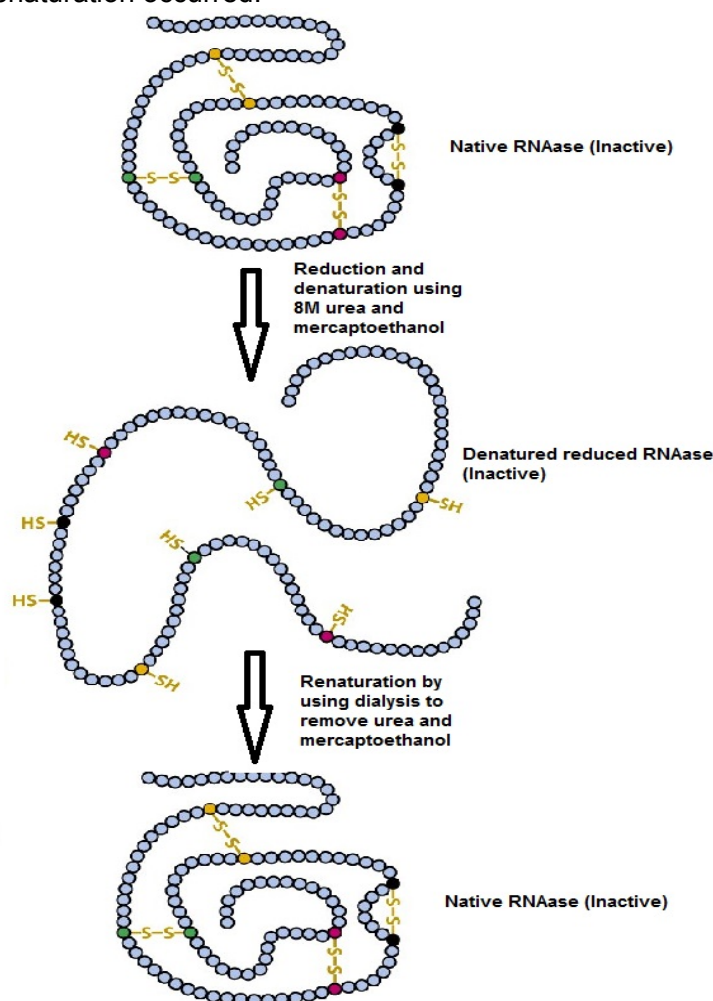
4. The diagram shows two molecules of glucose. Four possible bonding positions are labelled p, q, r and s on one molecule and t, u, v and w on another molecule.



Which of the following bonding positions are involved in the formation of glycogen?

- A p-u and p-v
- B p-u and q-w
- C p-v and q-w
- D p-w and v-w

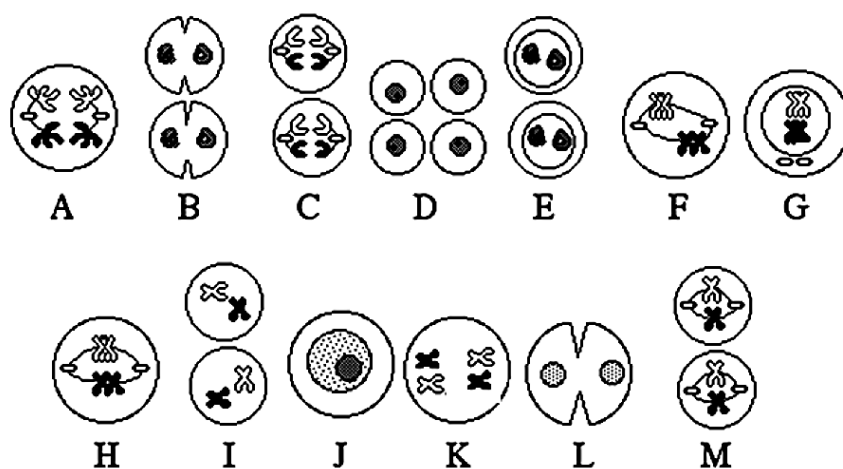
5. An experiment was carried out using pancreatic RNAase. A reducing agent, mercaptoethanol, and urea were added and denaturation was observed. Upon removal of the urea and mercaptoethanol, renaturation occurred.



Based on the above experiment, which of the following statements are true?

- I The polypeptide chain must have refolded as directed by its primary sequence and once refolded, the original disulphide bonds must have formed again.
 - II Given the right conditions, denatured proteins can refold.
 - III The tertiary structure of a protein is not determined by its primary structure.
 - IV Denaturation alters the primary structure of a protein and destroys its biological activity.
- A I and II only**
- B II and IV only**
- C II and III only**
- D I and IV only**

6. The diagram below shows cell division in a diploid organism.



Which of the following statements are correct?

- I The cell at stage J is in G2 phase.
- II Synapsis and crossing over occurs in the cell at stage K.
- III The order of sequence of events is $J \rightarrow G \rightarrow K \rightarrow H \rightarrow F \rightarrow A \rightarrow L \rightarrow I \rightarrow M \rightarrow C \rightarrow E \rightarrow B \rightarrow D$.
- IV The cell at stage F has the same number of chromosomes as a cell at stage C.

- A II and III only
- B I and IV only
- C I, II and IV
- D All of the above

7. The figures below show the complete karyotypes of 2 rodents of the same species. In this species of rodent, males are the heterogametic sex.



Rodent A (Male)



Rodent B (Female)

Which of the following observations is not true?

- A A chromosomal aberration occurred in the ovary of the mother of rodent **B**.
 - B Rodent **A** is diploid and $2n = 16$.
 - C Rodent **B** has 1 missing X chromosome.
 - D Non-disjunction of autosomes occurred in rodent **B**.
8. Polygalacturonase (PG) is expressed in tomato only during the ripening stage of fruit development. An anti-sense PG DNA construct was integrated into the tomato genome. The constitutive synthesis of PG anti-sense RNA in transgenic plants resulted in a substantial reduction in the levels of PG mRNA and enzymatic activity in ripening fruit.

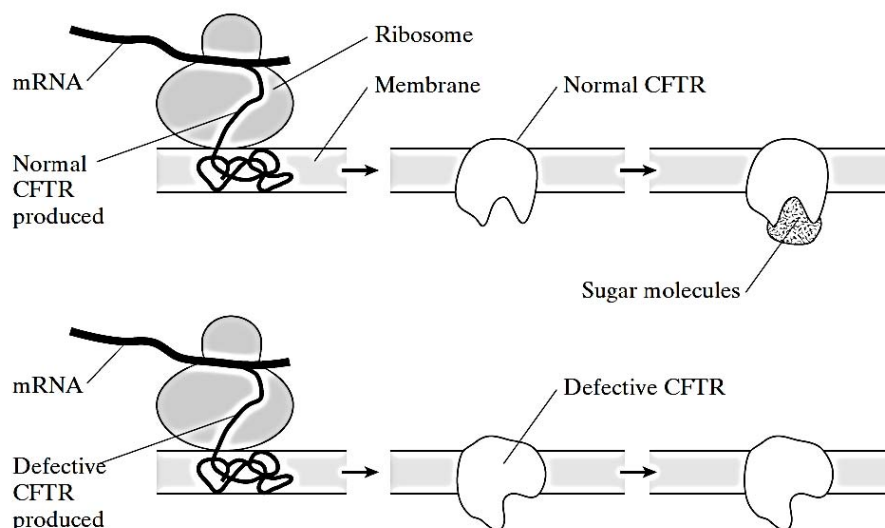
Part of the sequence of the template strand of the PG gene is shown below.

3' ... T T A G C C T T A ... 5'

Which of the following is the sequence of the anti-sense PG mRNA?

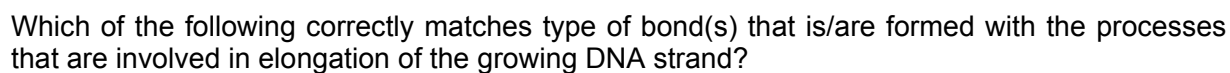
- A 5' – A A T C G G A A T – 3'
- B 5' – A U U C C G A U U – 3'
- C 5' – A A U C G G A A U – 3'
- D 5' – U U A G C C U U A – 3'

9. CFTR is a transmembrane regulator protein. It is made up of 1480 amino acids. People with cystic fibrosis produce a defective CFTR protein which is missing one amino acid from its structure. The diagram shows the synthesis of a normal and a defective CFTR in a cell. A normal CFTR protein molecule has sugar molecules attached to it which make it functional.



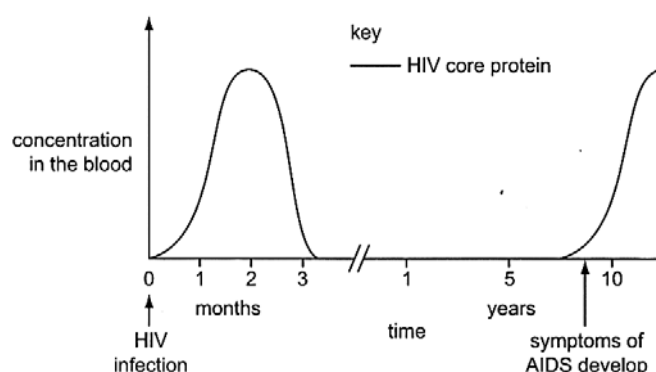
Which of the following statements are true?

- I 4440 is the number of bases on the template DNA which code for the amino acid sequence of a normal CFTR protein.
 - II Cystic fibrosis is due to a chromosomal mutation known as a deletion.
 - III The functional CFTR is a protein that undergone glycosylation.
 - IV The defective CFTR is not functional as the mutation in the DNA resulted in changes in its tertiary structure.
- A I and IV only
- B I and III only
- C I, III and IV only
- D II, III and IV only



	type of bond(s) formed	processes involved
A	phosphodiester and hydrogen	hydrolysis, polymerisation and complementary base pairing
B	phosphodiester	hydrolysis and condensation
C	phosphodiester and hydrogen	hydrolysis, condensation and complementary base pairing
D	hydrogen	complementary base pairing

11. The graph shows the changes in the core protein (capsid) concentration of the human immunodeficiency virus (HIV) in the blood stream after HIV infection. The protein is detected using antibodies.



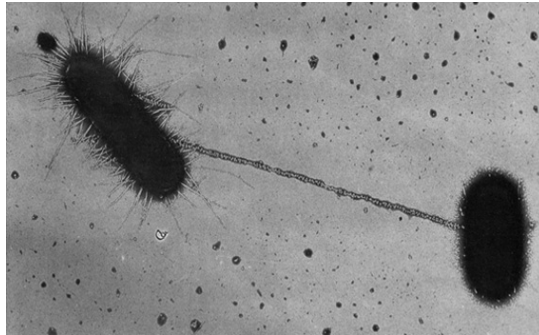
Which of the following explains the apparent absence of the protein between 3 months and 7 years?

- A The core protein attaches to a CD4 receptor so is not found in plasma.
 - B The lipid bilayer envelope masks the core protein.
 - C The patients' own antibodies destroyed the core protein.
 - D The virus enters the latent phase.
12. Temperate bacteriophages, such as the lambda phage, undergo the lysogenic cycle in their bacterial host cells.

Which of the following could prevent the temperate bacteriophages from entering the lysogenic life cycle?

- I A loss-of-function mutation in the viral gene which codes for integrase.
 - II Deletion of 10 nucleotides at the site of phage integration on the bacterial chromosome.
 - III Gain-of-function mutations in the viral genes which code for transcriptional repressor proteins.
 - IV Loss-of-function mutations in the viral genes which code for nucleases which break down bacterial chromosomal DNA.
- A I and III only
 - B I and II only
 - C II and IV only
 - D I, II and III only

13. The micrograph below shows the transfer of genetic material between bacterial cells by a bridge-like connection.



Which of the following statements about the above process are incorrect?

- I The transfer of the F plasmid is from the donor cell to recipient cell.
 - II For the transfer of plasmid DNA to occur, both strands of the double stranded F plasmid in the donor cell need to be nicked.
 - III After the sex pilus from the F^- cell attaches to the F^+ cells, it retracts.
 - IV Since DNA polymerase can only add nucleotides to the free 3'OH of a pre-existing strand, the end replication problem occurs during replication of the F plasmid.
 - V One strand of the F plasmid is transferred and the other remains in the original cell.
- A I and V only
 - B II and III only
 - C IV and V only
 - D II, III and IV only

14. X-gal is a compound which turns from colourless to blue, when cleaved by the enzyme beta-galactosidase. This enzyme is coded for by the *lac Z* structural gene, which is part of the *lac* operon in *E.coli*.

Which of the following mutations would result in the formation of blue colonies when *E.coli* cells are cultured in a medium containing X-gal but deficient in lactose and glucose?

- I A gain-of-function mutation in the gene coding for the catabolite activator protein.
- II A loss-of-function mutation in promoter region of the *lac I* regulatory gene.
- III A loss-of-function mutation which affects the DNA binding site of the Lac repressor
- IV A mutation which leads to the loss of the repressor binding site at the operator.
- V A mutation which leads to the loss of the cAMP binding sites of catabolite activator protein.

- A II and III only
- B I and V only
- C II, III and IV only
- D I, II, III and IV only

15. Which of the following occurs when a stem cell undergoes differentiation under the influence of molecular signals?

- A initiation of DNA replication
- B splicing of mature mRNA
- C base substitution in a silencer
- D production of activator molecule

16. Regulation of gene expression occurs in both prokaryotes and eukaryotes.

Which of the following process(es) is/are involved in gene regulation of both prokaryotes and eukaryotes?

- I extension of telomeres by telomerase
- II downregulation by repressor molecules
- III post-translational modification
- IV alternative splicing

- A II only
- B I and IV only
- C II and IV only
- D II, III and IV only

17. Which of the following is true of the eukaryotic genome?

- A There are certain chromosomes that may be inactivated.
- B DNA polymorphism is exhibited in different cell types of an organism.
- C Mutations in non-coding regions have no effect on selection.
- D In cancer cells, telomeres do not shorten as the end replication problem does not exist.

18. Which of the following processes is caused by an oncogene?

- A promotes the cell cycle
- B decrease rates of DNA replication
- C promotes DNA repair
- D controlled cell division

19. A population of cats contains individuals with shortened tails known as Manx cats and individuals with normal tails known as non-Manx cats. Mating of two non-Manx cats always produced non-Manx cats. Mating of two heterozygous Manx cats always produced a mixture of Manx cats and non-Manx cats, where the number of Manx cats is usually twice that of non-Manx cats.

Which hypothesis is consistent with these observations?

- A The genes for shortened tails and normal tails are linked on the same chromosome.
 - B The genes for shortened tails and normal tails are sex-linked.
 - C The allele for shortened tails is dominant over the allele for normal tails and the homozygous dominant genotype results in death.
 - D The allele for shortened tails is dominant over the allele for normal tails and the homozygous recessive genotype results in death.
20. In lentils, the seed coat pattern is determined by a gene with 3 alleles, C^M , C^S and C^C , whose phenotypes are marbled, spotted and clear respectively. Four crosses were repeated many times. The crosses and the outcomes of these crosses are shown in the table below.

cross	parents	offspring phenotype and ratio
1	marbled x marbled	3 marbled : 1 clear
2	spotted x clear	all spotted
3	marbled x marbled	3 marbled : 1 spotted
4	clear x clear	all clear

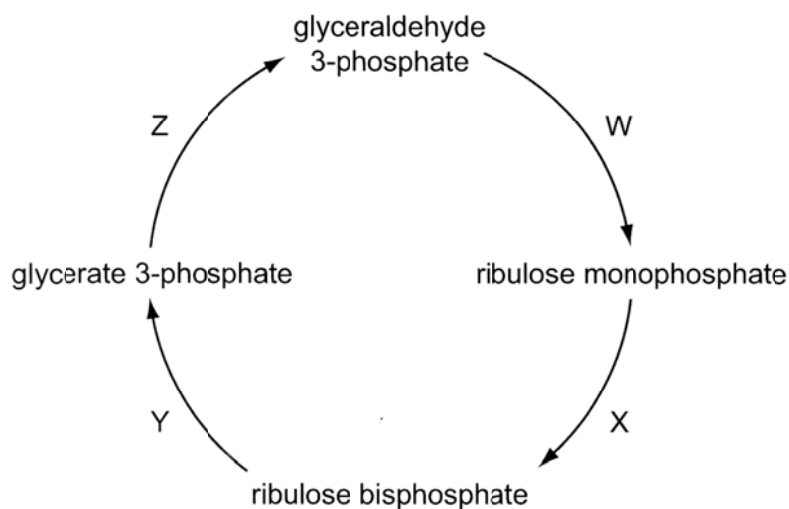
From the data, it is possible to conclude that,

- A spotted is recessive to clear.
- B all of the clear offspring are heterozygous.
- C two thirds of the marbled offspring in cross 3 are heterozygous.
- D the marbled parents in cross 1 have the same genotype as the marbled parents in cross 3.

21. *Cucurbita pepo* plants with the dominant allele W bear white fruit, whereas plants homozygous for the recessive allele w bear coloured fruit. When the fruit is coloured, the dominant allele B causes it to be yellow. In the absence of this allele (that is, with genotype bb), the fruit colour is green.

What is the expected F₂ phenotypic ratio from a cross of WWBB and wwbb plants? Assume that the W and B genes assort independently.

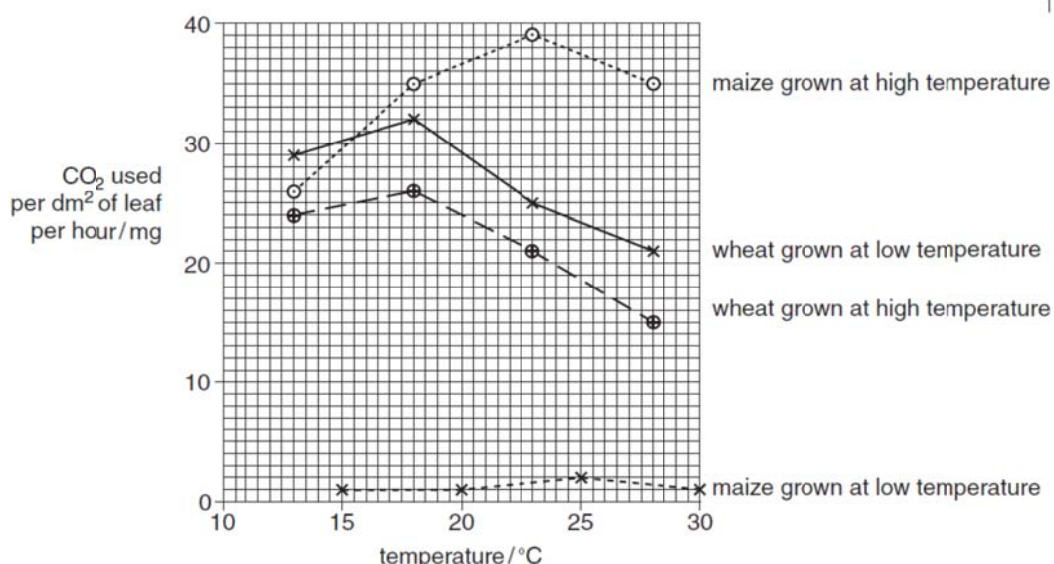
- A All white
 B 9 white: 7 yellow
 C 15 white: 1 green
 D 12 white: 3 yellow: 1 green
22. The diagram shows the main stages in the Calvin cycle.



At which stages are ATP and reduced NADP used and carbon dioxide taken up?

	ATP used	reduced NADP used	carbon dioxide taken up
A	W and Z	X	Y
B	X	W and Y	Z
C	Y	X and Z	W
D	X and Z	Z	Y

23. Young maize and wheat plants were grown to maturity at high and low temperatures. The rate of photosynthesis in each of these mature plants was measured at different temperatures. The rate of photosynthesis was measured as the amount of CO_2 used per dm^3 of leaf per hour. The results are shown in the graph below.



What information can be concluded from the graph above?

- I For plants grown at high temperature, the rate of photosynthesis is optimum at 25°C in maize and 18°C in wheat.
 - II For plants grown at high temperature, maize had a greater increase in rate of photosynthesis compared to wheat until optimum temperature was reached.
 - III The rate of photosynthesis was affected more significantly in maize plants than in wheat plants grown at low temperatures.
 - IV Low temperatures slowed down the formation of membranes in maize plants but not in wheat plants which caused a decrease in lamellae formation.
- A II and III only
 - B I and IV only
 - C I, II and III only
 - D All of the above

24. During electron transport, each reduced NAD produces on average 2.5 molecules of ATP and each reduced FAD produces on average 1.5 molecules of ATP.

Which of the following shows the correct number of ATP molecules produced in each stage of cellular respiration, from the oxidation of 1 molecule of glucose?

	molecules of ATP produced in each stage of cellular respiration			
	glycolysis	link reaction	Krebs cycle	electron transport
A	1	0	1	14
B	2	0	2	18
C	2	0	2	28
D	2	1	2	28

25. An inhibitor of the enzyme catalyzing the oxidative decarboxylation of α -ketoglutarate (5C) was added to respiring animal tissues.

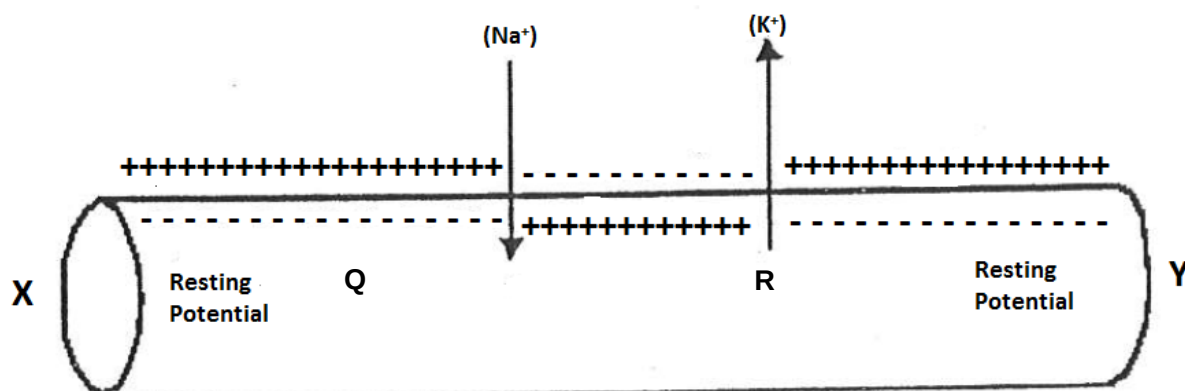
What would have been the effect of the inhibitor on the concentration of oxaloacetate, pyruvate and ATP?

	oxaloacetate	pyruvate	ATP
A	decreases	increases	decreases
B	decreases	decreases	increases
C	increases	increases	decreases
D	increases	decreases	increases

26. Which of the following identifies the role of acetylcholine in nerve physiology?

- A It increases the resting potential of the cell.
- B It increases the selective permeability of the cell membrane.
- C It activates the sodium pump mechanism of the cell.
- D It increases the potassium concentration inside the cell.

27. The diagram below shows a nerve impulse travelling along a nerve fibre. The arrows represent the direction of the flow of ions across the membrane.



Which of the following describes and explains the direction of the transmission of the impulse?

	impulse travelling towards	explanation
A	X	Region Q is more negative than region R.
B	X	Region R is in the refractory period.
C	Y	Region Q is in the refractory period.
D	Y	Sodium ions are moving towards region Q.

28. The mean levels of glucose and insulin in two groups of people are sampled and the results are shown in the table below.

	group P			group Q		
time after ingestion of 75g of glucose (min)	0	15	60	0	15	60
plasma glucose (mmol / L)	5.3	7.9	7.8	5.3	7.7	6.4
plasma insulin (pmol / L)	165	208	210	171	215	180

Which of the following statements about the above information is/are false?

- I People from group P lack beta-cells.
 - II Increased gluconeogenesis and glycogenolysis occur in people from group P after ingestion of 75g of glucose.
 - III More glucose can enter the target cells via facilitated diffusion in people from group Q due to the increase in numbers of the glucose transporters upon binding of insulin.
- A III only
 - B I and II only
 - C I and III only
 - D II and III only

29. Insulin receptors are involved in cell signalling. 4 different mutants, each with a single loss-of-function mutation affecting different domains of the receptor were produced.

Various functional abilities of the mutants and wild type were tested and the results were summarised in the table below. (The signs “+” and “-” indicate the presence and absence of the indicated functions respectively.)

receptor subtype	functional abilities			
	binding of insulin	conformational change of receptor	auto-phosphorylation	protein kinase activity
wild type	+	+	+	+
mutant 1	+	+	+	-
mutant 2	+	+	-	-
mutant 3	+	-	-	-
mutant 4	-	-	-	-

Based on the information given above, which of the following statement is false?

- A Receptors in mutant 2 lack tyrosine kinase activity.
- B Conformational changes in the receptor are necessary for autophosphorylation.
- C Protein kinase activity requires the binding of insulin.
- D Conformational changes of the receptor results in protein kinase activation.
30. The following describes certain structures found in two organisms.
- I fins of a dolphins and fish
- II legs of insects and lizards
- III wing bones of birds and bats
- IV appendix of the humans and cecum of the woolly monkey

Which of the above are homologous structures?

- A IV only
- B III and IV only
- C I and III only
- D I and II only

31. Which of the following is not biogeographical evidence supporting the descent from a common ancestor?
- A Mesosaur *Cynognathus* fossils have been found in both the continents of Africa and South America.
 - B Diverse early horse fossils have been found only in the continent of North America.
 - C The red fox, *Vulpes vulpes* is found in mainland Asia and Australia.
 - D Galapagos finches resemble the finches in Ecuador, South America.
32. Which of the following statements about natural selection is false?
- A Natural selection will result in a population acquiring new alleles that will confer advantageous phenotypes that will be selected for.
 - B Natural selection occurs even when the environmental conditions remain unchanged.
 - C Natural selection works to eliminate the unfit from the population through selection pressure, leaving the fit to reproduce.
 - D Natural selection works when the population is under a constant struggle for survival.
33. Listed below are some key concepts in the process of speciation.
- I natural selection
 - II changes in allele frequencies
 - III accumulation of unique mutations in sub-populations
 - IV isolation mechanism
 - V disruption of gene flow between populations
 - VI reproductive isolation

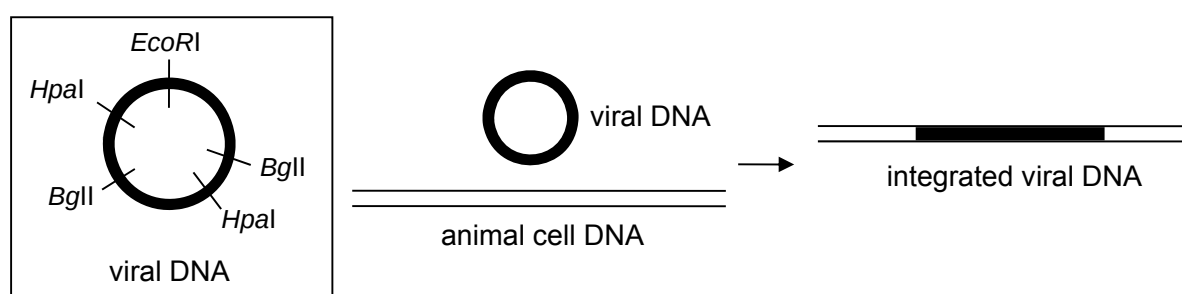
Which of the following shows the correct sequence of events that leads to speciation?

- A VI→V→IV→I→II→III
- B VI→IV→V→III→I→II
- C IV→V→VI→I→II→III
- D IV→V→I→II→III→VI

34. Which one of the following is not true about genomic and cDNA libraries?

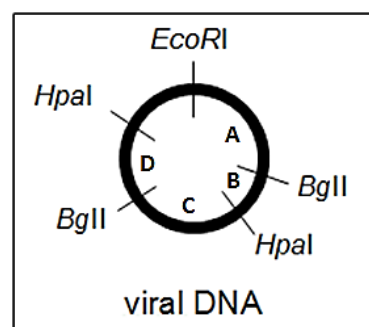
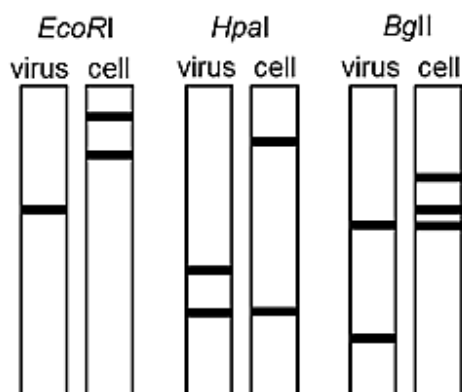
	genomic library	cDNA library
A	contains entire DNA content of an organism	contains entire protein-encoding DNA content of an organism
B	larger number of different clone types	smaller number of different clone types
C	cannot be used to study changes in gene expression levels	can be used to study gene expression at different stages of development
D	to make a genomic library, genetic material can be isolated from any somatic cell	to make a cDNA library, genetic material can only be isolated from a cell that produces the protein of interest

35. The DNA of a certain animal virus can integrate into its host cell's DNA as shown below. The restriction map of the viral DNA is also shown.



Using restriction endonucleases, a student cut the viral DNA and DNA from cells that contain the integrated viral DNA. The restriction fragments were separated by gel electrophoresis. Subsequently, the bands containing viral DNA were visualised by Southern blotting using radioactive probes (The probe binds to the entire region of the viral DNA.). The results of the Southern blots are shown below:

From the banding pattern, the student deduced that the integration event could have occurred in one of the 4 regions (labelled A, B, C or D) of the viral DNA.



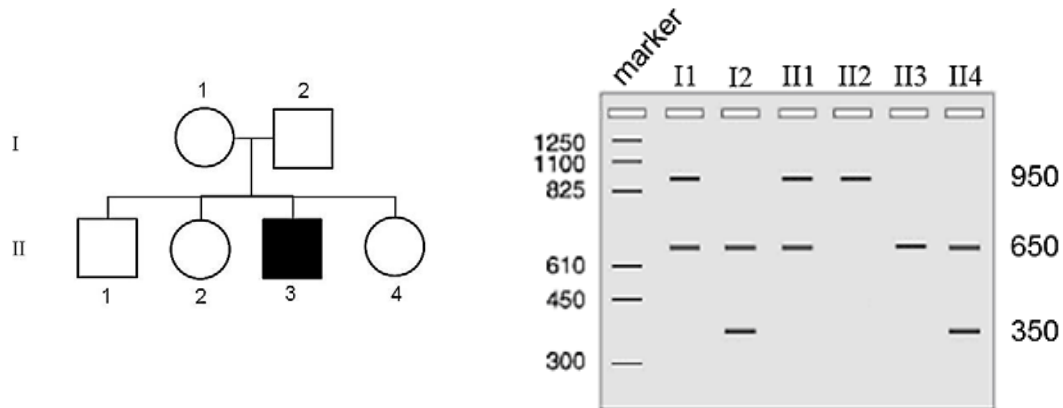
Which one of the following shows the correct region of integration of the viral DNA?

	region of integration	positions of restriction sites on integrated viral DNA
A	A	
B	B	
C	C	
D	D	

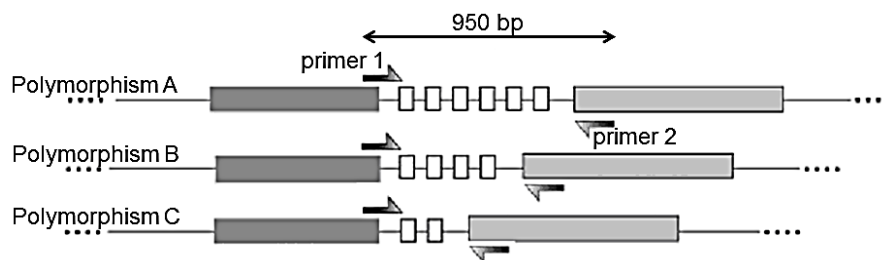
36. Which one of the following implications of the human genome project is not a concern?

- A Predictive genetic tests become available so that individuals can take steps to manage the disease.
- B If genetic sequences are patented, it will increase the cost of genetic research and treatment.
- C New genetic knowledge may conflict with a variety of ethical, social and legal perspectives.
- D DNA sequences of all individuals are readily accessible to the public.

37. In a study of a genetic disorder involving one gene, the DNA from six members of a family was isolated, amplified by PCR and run on an agarose gel. The pedigree was constructed using information obtained from interviewing family members.



The results reveal that there are three forms of the gene. Each gene has a different number of repeats in one of its introns.



Given the above information, which of the following statements is/are correct?

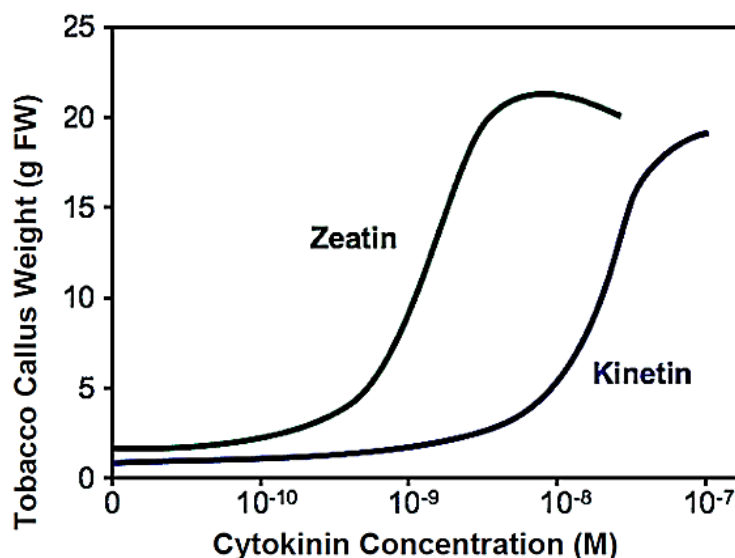
- I Polymorphism B marker is associated with the disease phenotype.
 - II It is possible, but rare, for individuals I1 and I2 to have an affected offspring with polymorphism B and C markers due to crossing over event during meiosis in individual I1.
 - III Individual I2 is not the biological father of individual II2.
 - IV The primers that flank the section of the DNA that is polymorphic are complementary to each other.
- A I and III only
- B I and II only
- C I, II and IV only
- D II, III and IV only

- 38.** Induced pluripotent stem cells are stem cells that can be generated directly from adult stem cells under the influence of molecular signals.

Which of the following statements are true?

- I** An induced pluripotent stem cell can become any cell of the developed organism, but cannot produce trophoblast and placenta to support organismal development, whereas a totipotent stem cell can produce a whole organism including extraembryonic tissue.
 - II** A totipotent stem cell and induced pluripotent stem cell can give rise to any cell.
 - III** An induced pluripotent stem cell can give rise to a single cell lineage whereas a totipotent stem cell can give rise to multiple, but limited number of cell lineages.
 - IV** A totipotent stem cell can become any cell of a developed organism, but cannot produce trophoblast and placenta to support organismal development, whereas an induced pluripotent stem cell can produce a whole organism including extraembryonic tissue.
 - V** Induced pluripotent stem cells have the same developmental potential as embryonic stem cells.
- A** I only
 - B** I and V only
 - C** II, III and IV only
 - D** III, IV and V only

39. The graph below shows the effect of natural cytokinin (zeatin) and synthetic cytokinin (kinetin) on tobacco callus growth.



Which of the following statements can be concluded from the graph?

- A Intermediate concentrations of kinetin and auxin were required for callus growth.
 - B Natural cytokinin is more effective at lower concentrations than artificial cytokinin in inducing tobacco callus growth.
 - C For callus growth, the optimum concentration for kinetin is 10^{-7} M and the optimum concentration for zeatin is 10^{-8} M.
 - D The tobacco callus growth increases with increasing zeatin concentration.
40. Transgenic corn which is capable of expressing a toxin found in *Bacillus thuringiensis* has been created by scientists.

Which one of the following is least likely to be affected by this genetically modified crop?

- A ratio of population sizes of different insect species within the farmland
- B growth of other crop plants within the farmland
- C quantity of insecticide required bought by the farmers each month
- D the population size of insects resistant to the toxin

End of Paper

Answers:

1.	A	11.	D	21.	D	31.	C
2.	C	12.	B	22.	D	32.	A
3.	C	13.	D	23.	A	33.	D
4.	B	14.	C	24.	C	34.	A
5.	A	15.	D	25.	A	35.	B
6.	B	16.	A	26.	B	36.	A
7.	D	17.	A	27.	B	37.	A
8.	B	18.	A	28.	B	38.	B
9.	C	19.	C	29.	D	39.	B
10.	A	20.	C	30.	B	40.	B